

Power BI Assignment 1 – Data Transformation & Data Modeling

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Module: Data Analytics (DA) – Module 2

Tool Used: Microsoft Power BI

1. Introduction

This project focuses on performing data transformation, cleaning, aggregation, merging, and data modeling using Power BI. The dataset contains E-commerce sales information including order details, product categories, customer locations, and sales targets.

2. Dataset Information

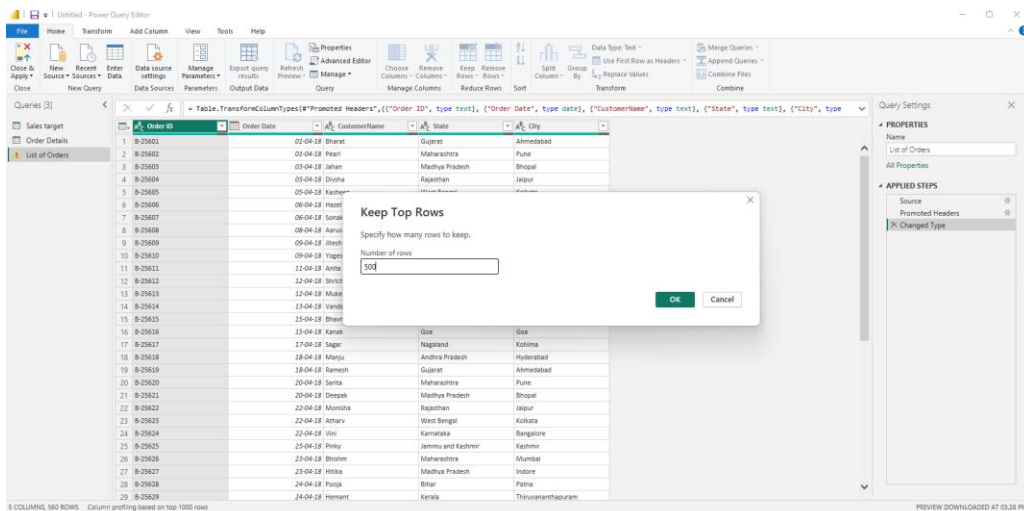
The following datasets were imported into Power BI:

- List of Orders
- Order Details
- Sales Target

3. Data Import & Transformation

The datasets were imported into Power BI and transformed using Power Query Editor. The following transformations were performed:

- Restricted List of Orders to first 500 rows
- Changed Order Date datatype to Date
- Changed Amount and Target columns to Fixed Decimal Number
- Formatted Customer Name into Proper Case
- Created Location column by merging City and State
- Created Profit Margin custom column
- Added Profit Status conditional column



Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group Use First Row By as Headers Count Rows

Table

Queries [3]

Sales target

Order Details

List of Orders

fx = Table.TransformColumns(#"Changed Type",{{"CustomerName",

Order ID Order Date CustomerName

1 B-25601 01-04-18 Bharat Ahmedabad

2 B-25602 01-04-18 Pearl Pune

3 B-25603 03-04-18 Jahan Bhopal

4 B-25604 03-04-18 Divsha Jaipur

5 B-25605 05-04-18 Kashem Kolkata

6 B-25606 06-04-18 Hazel Bangalore

7 B-25607 06-04-18 Sonakshi Jammu and Kashmir

8 B-25608 08-04-18 Asruthi Chennai

9 B-25609 09-04-18 Jitesh Lucknow

10 B-25610 09-04-18 Yogesh Patna

11 B-25611 11-04-18 Anita Thiruvananthapuram

12 B-25612 12-04-18 Shrichand Chandigarh

13 B-25613 12-04-18 Mukesh Haryana

14 B-25614 13-04-18 Vandana Himachal Pradesh

15 B-25615 15-04-18 Bhavna Gangtok

16 B-25616 15-04-18 Kanak Goa

17 B-25617 17-04-18 Sagar Kohima

18 B-25618 18-04-18 Manju Hyderabad

19 B-25619 18-04-18 Ramesh Ahmedabad

20 B-25620 20-04-18 Sanita Pune

21 B-25621 20-04-18 Deepak Bhopal

22 B-25622 22-04-18 Monisha Jaipur

23 B-25623 22-04-18 Atharv Kolkata

24 B-25624 22-04-18 Vini Bangalore

25 B-25625 23-04-18 Pinky Jammu and Kashmir

26 B-25626 23-04-18 Bhishm Mumbai

27 B-25627 23-04-18 Hrika Indore

28 B-25628 24-04-18 Pooja Patna

29 B-25629 24-04-18 Hemant Thiruvananthapuram

30 B-25630 24-04-18 Sahil Chandigarh

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group Use First Row By as Headers Count Rows

Table

Queries [3]

Sales target

Order Details

List of Orders

fx = Table.RenameColumns(#"Merged Columns",{{"Merged", "Location"}})

Order ID Order Date CustomerName Location

1 B-25601 01-04-18 Bharat Ahmedabad,Gujarat

2 B-25602 01-04-18 Pearl Pune,Maharashtra

3 B-25603 03-04-18 Jahan Bhopal,Madhya Pradesh

4 B-25604 03-04-18 Divsha Jaipur,Rajasthan

5 B-25605 05-04-18 Kashem Kolkata,West Bengal

6 B-25606 06-04-18 Hazel Bangalore,Karnataka

7 B-25607 06-04-18 Sonakshi Kashmir,Jammu and Kashmir

8 B-25608 08-04-18 Asruthi Chennai,Tamil Nadu

9 B-25609 09-04-18 Jitesh Lucknow,Uttar Pradesh

10 B-25610 09-04-18 Yogesh Patna,Bihar

11 B-25611 11-04-18 Anita Thiruvananthapuram,Kerala

12 B-25612 12-04-18 Shrichand Chandigarh,Punjab

13 B-25613 12-04-18 Mukesh Chandigarh,Haryana

14 B-25614 13-04-18 Vandana Simla,Himachal Pradesh

15 B-25615 15-04-18 Bhavna Gangtok,Sikkim

16 B-25616 15-04-18 Kanak Goa,Goa

17 B-25617 17-04-18 Sagar Kohima,Nagaland

18 B-25618 18-04-18 Manju Hyderabad,Andhra Pradesh

19 B-25619 18-04-18 Ramesh Ahmedabad,Gujarat

20 B-25620 20-04-18 Sanita Pune,Maharashtra

21 B-25621 20-04-18 Deepak Bhopal,Madhya Pradesh

22 B-25622 22-04-18 Monisha Jaipur,Rajasthan

23 B-25623 22-04-18 Atharv Kolkata,West Bengal

24 B-25624 22-04-18 Vini Bangalore,Karnataka

25 B-25625 23-04-18 Pinky Kashmir,Jammu and Kashmir

26 B-25626 23-04-18 Bhishm Mumbai,Maharashtra

27 B-25627 23-04-18 Hrika Indore,Madhya Pradesh

28 B-25628 24-04-18 Pooja Patna,Bihar

29 B-25629 24-04-18 Hemant Thiruvananthapuram,Kerala

30 B-25630 24-04-18 Sahil Chandigarh,Punjab

31 B-25631 24-04-18 Ritu Chandigarh,Haryana

32 B-25632 25-04-18 Manish Simla,Himachal Pradesh

33 B-25633 26-04-18 Amit Gangtok,Sikkim

34 B-25634 26-04-18 Sanjay Goa,Goa

35 B-25635 26-04-18 Nishi Kohima,Nagaland

36 B-25636 26-04-18 Nishi Mumbai,Maharashtra

37 B-25637 26-04-18 Ashmi Indore,Madhya Pradesh

38 B-25638 26-04-18 Parth Pune,Maharashtra

fx = Table.Distinct(#"Added Conditional Column")

Order ID Amount Profit Quantity Category Sub-Category % Profit Margin Profit status

1 B-25601 1,275.00 -1148 7 Furniture Bookcases -90.04% Loss

2 B-25601 66.00 -12 5 Clothing Stole -18.18% Loss

3 B-25601 8.00 -2 3 Clothing Hankerchief -25.00% Loss

4 B-25601 80.00 -56 4 Electronics Electronic Games -70.00% Loss

5 B-25602 168.00 -111 2 Electronics Phones -66.07% Loss

6 B-25602 424.00 -272 5 Electronics Phones -64.15% Loss

7 B-25602 2,617.00 1151 4 Electronics Phones 43.98% Profit

8 B-25602 561.00 212 3 Clothing Saree 37.79% Profit

9 B-25602 119.00 -5 8 Clothing Saree -4.20% Loss

10 B-25603 1,353.00 -60 5 Clothing Trousers -4.43% Loss

11 B-25603 24.00 -30 2 Furniture Chairs -125.00% Loss

12 B-25603 193.00 -166 3 Clothing Saree -86.01% Loss

13 B-25603 180.00 5 3 Clothing Trousers 2.78% Profit

14 B-25603 116.00 16 4 Clothing Stole 13.79% Profit

15 B-25603 107.00 36 6 Clothing Stole 33.64% Profit

16 B-25603 12.00 1 2 Clothing Hankerchief 8.33% Profit

17 B-25603 38.00 18 2 Clothing Kurti 47.37% Profit

18 B-25604 65.00 17 2 Clothing T-shirt 26.15% Profit

19 B-25604 157.00 5 9 Clothing Saree 3.18% Profit

20 B-25605 75.00 0 7 Clothing Saree 0.00% Break-even

21 B-25606 87.00 4 2 Clothing Shirt 4.60% Profit

22 B-25607 50.00 15 4 Clothing Leggings 30.00% Profit

23 B-25608 1,364.00 -1864 5 Furniture Tables -136.66% Loss

24 B-25608 476.00 0 3 Furniture Chairs 0.00% Break-even

25 B-25608 257.00 23 5 Clothing Hankerchief 8.95% Profit

26 B-25608 856.00 385 6 Electronics Printers 44.98% Profit

27 B-25609 485.00 29 4 Electronics Electronic Games 5.98% Profit

28 B-25609 25.00 -5 4 Clothing Saree -20.00% Loss

29 B-25610 1,076.00 -38 4 Electronics Printers -3.53% Loss

30 B-25610 107.00 -54 4 Clothing Stole -50.47% Loss

31 B-25610 68.00 -55 3 Electronics Accessories -80.88% Loss

32 B-25610 781.00 -594 6 Electronics Printers -76.06% Loss

33 B-25610 43.00 0 3 Clothing Saree 0.00% Break-even

34 B-25610 30.00 -5 2 Furniture Furnishings -16.67% Loss

35 B-25611 160.00 -59 2 Clothing Saree -36.88% Loss

36 B-25612 259.00 -55 2 Furniture Chairs -21.24% Loss

37 B-25613 1,603.00 0 9 Clothing Saree 0.00% Break-even

38 B-25614 494.00 54 4 Furniture Bookcases 10.93% Profit

4. Data Merging

List of Orders and Order Details tables were merged using Order ID to create a consolidated table named Orders Data.

fx

=Table.SelectRows("#Sorted Rows", each true)

	Date	CustomerName	Location	Order Details.Amount	Order Details.Profit	Order Details.Quantity	Order Details.Category	Order Details.Sub-Category	Order Details.Profit Margin	Order Details.Profit status
1	30-03-19	Bhishm	Mumbai,Maharashtra	9.00	3	1	Clothing	Skirt	33.33%	Profit
2	30-03-19	Bhishm	Mumbai,Maharashtra	2,366.00	552	5	Clothing	Trousers	23.33%	Profit
3	30-03-19	Bhishm	Mumbai,Maharashtra	207.00	37	4	Clothing	Hankerchief	17.87%	Profit
4	30-03-19	Bhishm	Mumbai,Maharashtra	835.00	267	5	Electronics	Phones	31.98%	Profit
5	29-03-19	Pinky	Kashmir,Jammu and Kashmir	409.00	86	3	Clothing	Saree	21.03%	Profit
6	29-03-19	Pinky	Kashmir,Jammu and Kashmir	497.00	179	3	Furniture	Chairs	36.02%	Profit
7	29-03-19	Pinky	Kashmir,Jammu and Kashmir	96.00	48	5	Clothing	Leggings	50.00%	Profit
8	29-03-19	Pinky	Kashmir,Jammu and Kashmir	82.00	8	3	Electronics	Accessories	9.76%	Profit
9	29-03-19	Pinky	Kashmir,Jammu and Kashmir	59.00	15	2	Clothing	T-shirt	25.42%	Profit
10	29-03-19	Pinky	Kashmir,Jammu and Kashmir	46.00	14	5	Clothing	Skirt	30.43%	Profit
11	28-03-19	Vini	Bangalore,Karnataka	19.00	8	2	Clothing	Hankerchief	42.11%	Profit
12	28-03-19	Atharv	Kolkata,West Bengal	264.00	-26	3	Clothing	Trousers	-9.85%	Loss
13	28-03-19	Vini	Bangalore,Karnataka	663.00	-212	5	Electronics	Printers	-31.98%	Loss
14	28-03-19	Vini	Bangalore,Karnataka	185.00	-26	6	Furniture	Chairs	-14.05%	Loss
15	28-03-19	Atharv	Kolkata,West Bengal	88.00	11	3	Electronics	Accessories	12.50%	Profit
16	28-03-19	Vini	Bangalore,Karnataka	39.00	18	2	Clothing	Leggings	46.15%	Profit
17	28-03-19	Vini	Bangalore,Karnataka	671.00	-309	5	Electronics	Electronic Games	-46.05%	Loss
18	28-03-19	Monisha	Jaipur,Rajasthan	6.00	1	1	Clothing	Kurti	16.67%	Profit
19	28-03-19	Vini	Bangalore,Karnataka	57.00	12	2	Clothing	Hankerchief	12.37%	Profit
20	28-03-19	Atharv	Kolkata,West Bengal	451.00	25	3	Electronics	Phones	5.54%	Profit
21	28-03-19	Vini	Bangalore,Karnataka	14.00	5	1	Clothing	Hankerchief	35.71%	Profit
22	28-03-19	Atharv	Kolkata,West Bengal	140.00	56	4	Clothing	Shirt	40.00%	Profit
23	28-03-19	Atharv	Kolkata,West Bengal	103.00	46	2	Clothing	Saree	44.66%	Profit
24	28-03-19	Atharv	Kolkata,West Bengal	45.00	9	3	Clothing	Leggings	20.00%	Profit
25	27-03-19	Sarita	Pune,Maharashtra	492.00	187	2	Electronics	Accessories	38.01%	Profit
26	27-03-19	Sarita	Pune,Maharashtra	2,847.00	712	8	Electronics	Printers	25.01%	Profit
27	27-03-19	Sarita	Pune,Maharashtra	81.00	41	5	Clothing	Leggings	50.62%	Profit
28	27-03-19	Manju	Hyderabad,Andhra Pradesh	59.00	10	4	Clothing	Leggings	16.95%	Profit
29	27-03-19	Sarita	Pune,Maharashtra	33.00	-1	1	Clothing	Saree	-3.03%	Loss
30	27-03-19	Sarita	Pune,Maharashtra	49.00	5	4	Clothing	Hankerchief	10.20%	Profit
31	27-03-19	Sarita	Pune,Maharashtra	148.00	25	3	Clothing	Saree	16.89%	Profit
32	27-03-19	Manju	Hyderabad,Andhra Pradesh	29.00	10	4	Clothing	Hankerchief	34.48%	Profit
33	27-03-19	Deepak	Bhopal,Madhya Pradesh	152.00	50	6	Clothing	Stole	32.89%	Profit
34	27-03-19	Manju	Hyderabad,Andhra Pradesh	158.00	69	3	Clothing	Stole	43.67%	Profit
35	27-03-19	Ramesh	Ahmedabad,Gujarat	97.00	14	2	Clothing	T-shirt	14.43%	Profit
36	27-03-19	Sagar	Kohima,Nagaland	80.00	22	3	Clothing	Stole	27.50%	Profit
37	27-03-19	Sarita	Pune,Maharashtra	852.00	51	5	Furniture	Bookcases	5.99%	Profit
38	27-03-19	Sarita	Pune,Maharashtra	130.00	14	3	Clothing	Kurti	10.77%	Profit

5. Missing Values & Duplicate Handling

The datasets were checked for missing values and duplicate records. Appropriate handling strategies such as retaining valid business records and removing unnecessary duplicates were applied.

6. Sorting & Filtering

The Orders Data table was sorted by Order Date in descending order. Filters were applied to analyze orders from specific states such as Tamil Nadu.

Column query

Column

Editor

Dependencies

Data Preview

Columns

Parameters

Advanced

Dependencies

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✕

✓

f_x

= Table.SelectRows (#"Sorted Rows", each ([Location] = "Chennai,Tamil Nadu"))

⌵

A ₀	Order ID	Order Date	A ₀	CustomerName	A ₀	Location	\$	Order Details.Amount	1 ₃	Order Details.Profit	1 ₃	Order Details.Quantity	A ₀	Order Details.Category	A ₀	Order Data
1	B-26081	22-03-19	Aarushi	Chennai,Tamil Nadu			79.00		33			4	Clothing		Stole	
2	B-26081	22-03-19	Aarushi	Chennai,Tamil Nadu			169.00		0			3	Electronics		Accessories	
3	B-26081	22-03-19	Aarushi	Chennai,Tamil Nadu			359.00		-338			5	Furniture		Bookcases	
4	B-26081	22-03-19	Aarushi	Chennai,Tamil Nadu			93.00		-84			3	Clothing		Saree	
5	B-26081	22-03-19	Aarushi	Chennai,Tamil Nadu			24.00		11			3	Clothing		Hankerchief	
6	B-26081	22-03-19	Aarushi	Chennai,Tamil Nadu			637.00		50			5	Clothing		Saree	
7	B-26018	14-02-19	Aarushi	Chennai,Tamil Nadu			61.00		8			4	Clothing		Hankerchief	
8	B-26018	14-02-19	Aarushi	Chennai,Tamil Nadu			326.00		107			3	Furniture		Furnishings	
9	B-26008	09-02-19	Kavyani	Chennai,Tamil Nadu			22.00		4			1	Clothing		Stole	
10	B-26008	09-02-19	Kavyani	Chennai,Tamil Nadu			206.00		51			4	Clothing		Hankerchief	
11	B-26008	09-02-19	Kavyani	Chennai,Tamil Nadu			10.00		-1			1	Clothing		Saree	
12	B-26008	09-02-19	Kavyani	Chennai,Tamil Nadu			57.00		24			5	Clothing		Leggings	
13	B-25860	15-11-18	Akshay	Chennai,Tamil Nadu			112.00		24			3	Clothing		Kurti	
14	B-25788	21-09-18	Dinesh	Chennai,Tamil Nadu			12.00		3			1	Clothing		Stole	
15	B-25716	11-07-18	Surabhi	Chennai,Tamil Nadu			58.00		0			4	Clothing		Saree	
16	B-25698	23-06-18	Amisha	Chennai,Tamil Nadu			207.00		-153			3	Clothing		Saree	
17	B-25698	23-06-18	Amisha	Chennai,Tamil Nadu			27.00		-6			4	Clothing		Hankerchief	
18	B-25698	23-06-18	Amisha	Chennai,Tamil Nadu			65.00		-16			2	Electronics		Phones	
19	B-25698	23-06-18	Amisha	Chennai,Tamil Nadu			87.00		-83			5	Clothing		Kurti	
20	B-25698	23-06-18	Amisha	Chennai,Tamil Nadu			7.00		-2			1	Clothing		Hankerchief	
21	B-25698	23-06-18	Amisha	Chennai,Tamil Nadu			516.00		-392			8	Furniture		Chairs	
22	B-25608	08-04-18	Aarushi	Chennai,Tamil Nadu			257.00		23			5	Clothing		Hankerchief	
23	B-25608	08-04-18	Aarushi	Chennai,Tamil Nadu			856.00		385			6	Electronics		Printers	
24	B-25608	08-04-18	Aarushi	Chennai,Tamil Nadu			476.00		0			3	Furniture		Chairs	
25	B-25608	08-04-18	Aarushi	Chennai,Tamil Nadu			1,364.00		-1864			5	Furniture		Tables	

PROPERTIES

Name

Order Data

All Properties

APPLIED STEPS

Source

Expanded Order Details

Removed Duplicates

Sorted Rows

Filtered Rows

7. Grouping & Aggregation

Aggregated tables were created using duplicate queries:

- Count of Order ID
- Average Profit by Category
- Total Amount by Sub-Category
- Total Target by Month of Order Date

	Data Sources	Parameters	Output Data	Query	Manage Columns	Reduce Rows	Sort
n Query				= Table.Group("#Removed Duplicates", {"Category"}, {{ "Avg Profit By category", each Li			
<	X ✓ f _x						
	AB C Category		1.2 Avg Profit By category				
1	Furniture		9.456790123				
2	Clothing		11.76290832				
3	Electronics		34.07142857				

	Order ID	Count of OrderID
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2
10	B-25610	6
11	B-25611	1
12	B-25612	1
13	B-25613	1
14	B-25614	2
15	B-25615	1
16	B-25616	4
17	B-25617	1
18	B-25618	2
19	B-25619	1
20	B-25620	1
21	B-25621	3
22	B-25622	1
23	B-25623	4
24	B-25624	1
25	B-25625	3
26	B-25626	2
27	B-25627	1
28	B-25628	3

	Month of Order Date	1.2 Total target
1	01-04-18	31400
2	01-05-18	31500
3	01-06-18	31600
4	01-07-18	33800
5	01-08-18	33900
6	01-09-18	34000
7	01-10-18	36100
8	01-11-18	36300
9	01-12-18	36400
10	01-01-19	43500
11	01-02-19	43600
12	01-03-19	43800

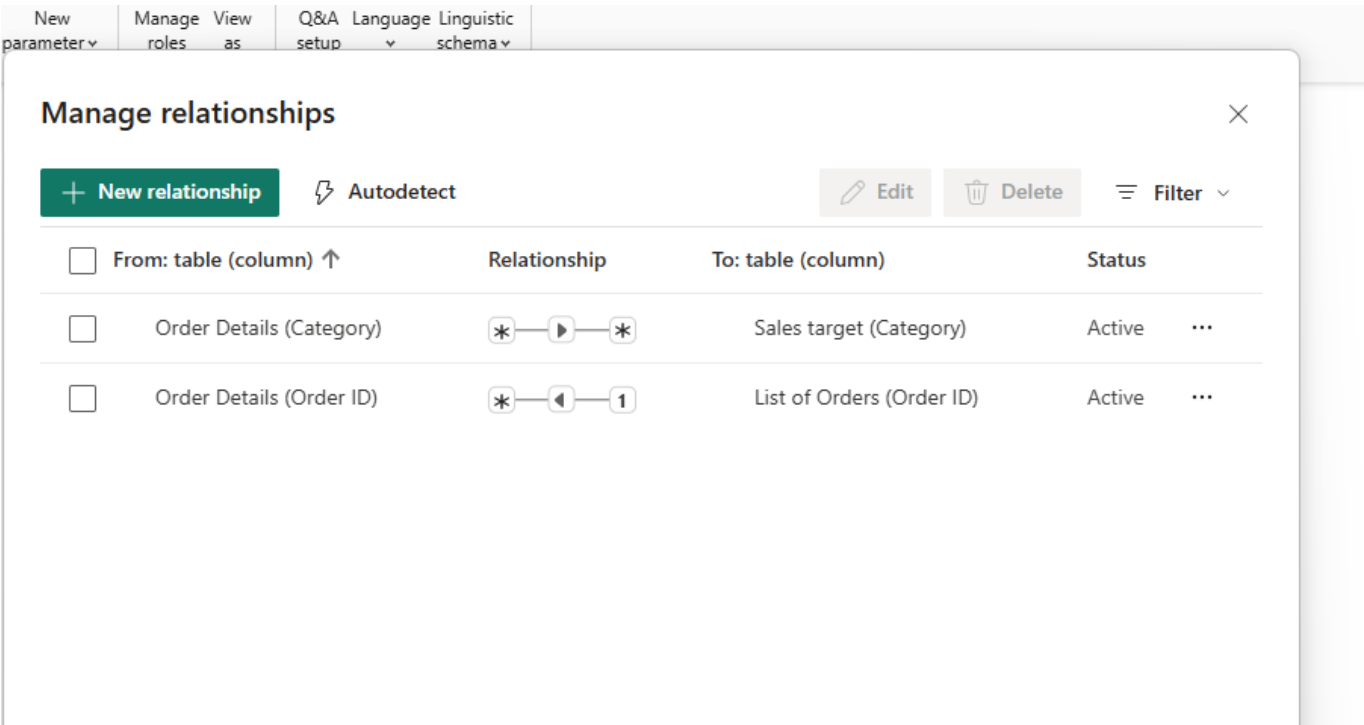
	Month of Order Date	1.2 Total target
1	01-04-18	31400
2	01-05-18	31500
3	01-06-18	31600
4	01-07-18	33800
5	01-08-18	33900
6	01-09-18	34000
7	01-10-18	36100
8	01-11-18	36300
9	01-12-18	36400
10	01-01-19	43500
11	01-02-19	43600
12	01-03-19	43800

8. Data Modeling

Relationships were created and managed in Model View:

- List of Orders ↔ Order Details using Order ID
- Order Details ↔ Sales Target using Category

Relationships were active and correctly configured using Manage Relationships.



9. Conclusion

This project demonstrates practical knowledge of Power BI data transformation, aggregation, data integration, and relationship modeling techniques. The final data model was cleaned, structured, and prepared for business analysis.